CORREÇÃO - 2ª PROVA

1)

"Função contínua num ponto a do domínio"

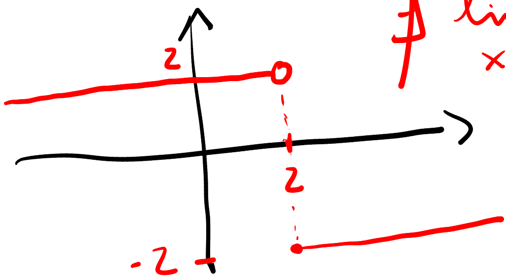
$$\begin{cases} a \in D_f \\ \underbrace{\exists}_{\text{EXISTE}} \lim\limits_{x \to a} f(x) \\ \lim\limits_{x \to a} f(x) = f(a) \end{cases}$$

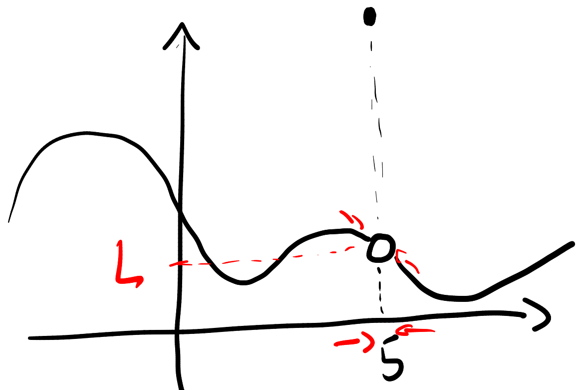
"Função contínua"

$$\begin{cases} \text{É a função contínua} \\ \text{em todos os pontos} \\ \text{do } D_f \end{cases}$$

EXEMPLOS:

$\underbrace{\nexists}_{\text{NÃO EXISTE}} \lim\limits_{x \to 2} f(x)$





$$\lim_{x \to 5} f(x) = L \neq f(5)$$

2)

$$a^2 - b^2 = (a+b)(a-b)$$

a)

$$\lim_{x \to 16} \frac{\overset{(\sqrt{x})^2}{x} - \overset{(4)^2}{16}}{\underbrace{\sqrt{x} - 4}}$$

$x \to 16 \Rightarrow x \approx 16 \Rightarrow$

$$f(x) \approx \frac{16 - 16}{\sqrt{16} - 4} \approx \frac{0}{0} \ (!?) \quad \text{INDET.}$$

$$= \frac{(\sqrt{x})^2 - (4)^2}{\sqrt{x} - 4} = \frac{(\sqrt{x} + 4).\cancel{(\sqrt{x} - 4)}}{\cancel{(\sqrt{x} - 4)}} = \sqrt{x} + 4$$

$$\lim_{x\to 16} f(x) = \lim_{x\to 16} (\sqrt{x}+4) = \sqrt{16}+4$$
$$= 4+4 = 8$$

b)

$(x-(-2)) = (x+2)$

-2 é raiz $\Rightarrow$ $(\underbrace{x - \text{raiz}}_{(x-(-2))=(x+2)})$ divide x^3+8

$$\lim_{x\to -2} \frac{x^3+8}{x^4-16}$$

x^4-16: -2 é raiz

$(x^2)^2 - (4)^2$

$= (x^2+4).\underbrace{(x^2-4)}_{(x+2)(x-2)}$

$$\begin{array}{l|l} x^3+0.x^2+0.x+8 & x+2 \\ \hline -x^3-2x^2 & x^2-2x+4 \\ -2x^2+0x & \\ +2x^2+4x & \\ 4x+8 & \\ -4x-8 & \\ 0 & \end{array}$$

Logo, $x^4-16 = (x^2+4).(x+2).(x-2)$

$x^3+8 = (x+2).(x^2-2x+4)$

$$f(x) = \frac{x^3+8}{x^4-16} = \frac{\cancel{(x+2)}.(x^2-2x+4)}{(x^2+4).\cancel{(x+2)}.(x-2)} = \frac{x^2-2x+4}{(x^2+4).(x-2)}$$

$$\lim_{x\to -2} f(x) = \frac{(-2)^2-2.(-2)+4}{((-2)^2+4).(-2-2)} = \frac{4+4+4}{8.(-4)} = \frac{3.\cancel{4}}{8.-\cancel{4}}$$

$$= -3/8$$

c)

$$\lim_{h\to 0}\frac{(9+h)^{-1}-9^{-1}}{h} = \frac{\frac{1}{9+h} - \frac{1}{9}}{h}$$

$9 \times$ $\times(9+h)$

$$= \frac{\frac{\cancel{9} - \overbrace{(\cancel{9}+h)}^{-9-h}}{9.(9+h)}}{h} = \frac{\boxed{\frac{-h}{9.(9+h)}}}{\frac{h}{1}} = \frac{-h}{9.(9+h)} \cdot \frac{1}{h}$$

1ª fração

2ª fração

$$= \frac{-\cancel{h}}{9.\cancel{h}.(9+h)} = \frac{-1}{9.(9+h)}$$

$$\lim_{h\to 0} f(h) = \frac{-1}{9.(9+0)} = -\frac{1}{81}$$

$$\boxed{\lim_{h\to 0}\frac{(2+h)^{-2}-2^{-2}}{h}} \quad f(x)$$

obs: $x^{-2} = 1/x^2$

$4 \times$ $\times(2+h)^2$

$$f(x) = \frac{\boxed{\frac{1}{(2+h)^2} - \frac{1}{2^2}}}{h} = \frac{\frac{4 - (2+h)^2}{4.(2+h)^2}}{\frac{h}{1}}$$

$$= \frac{\frac{\cancel{4} - (\cancel{4}+4h+h^2)}{4.(2+h)^2}}{h} = \frac{-4h-h^2}{4.(2+h)^2} \cdot \frac{1}{h} = \frac{\cancel{h}(-4-h)}{4.(2+h)^2.\cancel{h}}$$

$$= \frac{-4-h}{4.(2+h)^2}$$

$$\lim_{h \to 0} f(h) = \frac{-4}{4.(2)^2} = \frac{-4}{16} = -1/4$$

e)

2 é raiz

$$\lim_{x\to 2} \frac{(x-2)^1}{x^3-8} = \lim_{x \to 2} \frac{(x-2)}{x^3-8} = \frac{0}{0} \quad (!?)$$

2 é raiz

↳ $(x-2)$ divide x^3-8

$$\begin{array}{l|l}
\cancel{x^3} - 8 & x-2 \\
\hline
-\cancel{x^3} + 2x^2 & x^2+2x+4 \\
\hline
\cancel{2x^2} & \\
-\cancel{2x^2} + 4x & \\
\hline
\cancel{4x} - \cancel{8} & \\
-\cancel{4x} + \cancel{8} & \\
\hline
0 &
\end{array}$$

$\Rightarrow x^3 - 8 = (x-2).(x^2+2x+4)$

$$\lim_{x\to 2} \frac{x-2}{x^3-8} = \lim_{x \to 2} \frac{\cancel{(x-2)}}{\cancel{(x-2)}.(x^2+2x+4)} = \lim_{x\to 2} \frac{1}{x^2+2x+4}$$

$$= \frac{1}{4+4+4} = 1/12$$

f)

$$\lim_{h\to 0}\frac{\frac{1}{\sqrt{1+h}}-1}{h}$$

Obs: Note que tal limite é a derivada da $1/\sqrt{x}$ no ponto $x = 1$.

$$\frac{\frac{1}{\sqrt{1+h}}-1}{h} = \frac{\frac{1-\sqrt{1+h}}{\sqrt{1+h}}}{h}\cdot\frac{1+\sqrt{1+h}}{1+\sqrt{1+h}}$$

$$= \frac{\frac{\cancel{1}+\cancel{\sqrt{1+h}}-\cancel{\sqrt{1+h}}-(\cancel{1}+h)}{\sqrt{1+h}\,(1+\sqrt{1+h})}}{h} = \frac{-\cancel{h}}{\sqrt{1+h}\,(1+\sqrt{1+h})}\cdot\frac{1}{\cancel{h}}$$

$$= \frac{-1}{\sqrt{1+h}\cdot(1+\sqrt{1+h})} \Rightarrow \lim_{h\to 0} f(h) = \frac{-1}{\sqrt{1}\cdot(\underbrace{1+\sqrt{1}}_{2})} = \frac{-1}{2}$$

g)

$$\lim_{x\to -10^+}\frac{x+10}{|x+10|}$$

$x+10$ $-(x+10)$

$$= \lim_{x\to -10^+}\frac{x+10}{x+10} = \lim_{x\to -10^+}(1) = 1$$

$x \to -10^+ \Rightarrow x > -10 \Rightarrow x+10 > 0$

$\Downarrow$

$|\underbrace{x+10}_{>0}| = x+10$

h)

$$\lim_{x\to 4+}\left(\frac{\sqrt[12]{x^2-16}}{|x+4|}+\frac{1}{x-4}\right)$$

$$= \lim_{x\to 4^+}\left(\frac{\sqrt[12]{x^2-16}}{|x+4|}\right) + \underbrace{\lim_{x\to 4^+}\left(\frac{1}{x-4}\right)}_{+\infty} \quad \frac{1}{0^+}$$

$$= \underbrace{\frac{\sqrt[12]{(4^+)^2-16}}{|4^+ + 4|} = \frac{0^+}{8^+}}_{0}$$

$$0 + \infty = +\infty$$